

Ellipses and Hyperbolas in Real Structures

Answer Key

Algebra 2

Quick Answers

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| <p>1. (a) the major axis = 26 m, (b) the minor axis = 24 m</p> <p>2. 10 m</p> <p>3. $6\sqrt{3}$ ft</p> <p>4. (a) the vertex separation = 16 m, (b) the focal distance = 10 m</p> <p>5. (a) $= \frac{3}{4}$, —</p> <p>6. (a) the focal distance = 8 m, (b) the</p> | <p>constant sum = 20 m</p> <p>7. (a) the vertex distance = 3 cm, (b) the focal distance = 5 cm</p> <p>8. (a) the asymptote slope = $\frac{5}{12}$, (b) the vertex distance = 5 cm</p> <p>9. 24 ft</p> <p>10. (a) = 12 m, —</p> |
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What is verified

15 of 17 answers machine-checked against SymPy, across 10 problems.
 — marks an answer only you can judge — problems 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSG-GPE.A.1–A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 17 tagged checks.

- 1.** Whispering gallery floor $\frac{x^2}{169} + \frac{y^2}{144} = 1$ (metres). (a) Major axis. (b) Minor axis.
Solution (a): The denominators are a^2 and b^2 . The larger, 169, sits under x^2 , so the long axis is horizontal with half-length $\sqrt{169} = 13$; the full major axis is twice that. 26 m
Solution (b): The half-length of the short axis is $\sqrt{144} = 12$, so the minor axis is 24. 24 m
 The room is 26 metres across at its widest and 24 metres across at its narrowest.
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- 2.** Distance between the two foci of $\frac{x^2}{169} + \frac{y^2}{144} = 1$ (metres).
Solution: For an ellipse the foci sit inside, so $c^2 = a^2 - b^2$ — subtract.

$$c = \sqrt{169 - 144} = \sqrt{25} = 5$$

The two listening spots are $2c$ apart.

10 m

3. Height of the arch $\frac{x^2}{400} + \frac{y^2}{144} = 1$ at $x = 10$ (feet).

Solution: Solve the equation for y and substitute $x = 10$; the arch is the upper half, so take the positive root.

$$\begin{aligned}\frac{y^2}{144} &= 1 - \frac{100}{400} = \frac{3}{4} \\ y^2 &= 108 \\ y &= \sqrt{108} = 6\sqrt{3}\end{aligned}$$

$$6\sqrt{3} \text{ ft}$$

That is about 10.4 feet, comfortably below the 12-foot centre height, as it must be.

4. Cooling tower $\frac{x^2}{64} - \frac{y^2}{36} = 1$ (metres). (a) Vertex separation. (b) Focal distance.

Solution (a): The x^2 term is the positive one, so the branches open left and right and the vertices sit at $\pm\sqrt{64} = \pm 8$. The distance across the waist is twice that. 16 m

Solution (b): For a hyperbola $c^2 = a^2 + b^2$ — add, the opposite of the ellipse rule, because the foci sit beyond the vertices.

$$c = \sqrt{64 + 36} = \sqrt{100}$$

$$10 \text{ m}$$

5. Asymptotes of $\frac{x^2}{64} - \frac{y^2}{36} = 1$. (a) Rising slope. (b) What they describe.

Solution (a): For this orientation the asymptotes are $y = \pm \frac{b}{a}x$ with $a = \sqrt{64} = 8$ and $b = \sqrt{36} = 6$.

$$\frac{6}{8}$$

The rising asymptote has slope

$$\frac{3}{4}$$

Solution (b) — instructor-judged. High above the waist the tower wall follows the asymptotes almost exactly, so the silhouette there is effectively two straight lines of slope $\pm \frac{3}{4}$ — but the wall never actually reaches them; the gap keeps shrinking without ever closing. Full credit says both things: the wall gets arbitrarily close, and it never touches. Describing the asymptotes as the edges of the tower, or as a line the wall meets, is the misconception this part exists to catch.

6. Planetarium dome $\frac{x^2}{100} + \frac{y^2}{36} = 1$ (metres). (a) Focal distance. (b) Cable length.

Solution (a): $c^2 = a^2 - b^2$ for an ellipse:

$$c = \sqrt{100 - 36} = \sqrt{64}$$

$$8 \text{ m}$$

Solution (b): The defining property of an ellipse is that the two focal distances add to the same total everywhere on the curve, and that total is the major axis $2a$. Checking at a vertex: the near focus is $a - c = 2$ away and the far one $a + c = 18$, totalling 20. 20 m

7. Hyperbolic mirror $\frac{y^2}{9} - \frac{x^2}{16} = 1$ (centimetres). (a) Vertex distance. (b) Focal distance.

Solution (a): The y^2 term is positive, so the branches open up and down and the vertices lie $\sqrt{9}$ from the centre. 3 cm

Solution (b): Adding, as a hyperbola requires:

$$c = \sqrt{9 + 16} = \sqrt{25}$$

5 cm

The focus is further from the centre than the vertex, which is always true for a hyperbola and never for an ellipse.

8. Shadow line $\frac{y^2}{25} - \frac{x^2}{144} = 1$ (centimetres). (a) Rising asymptote slope. (b) Vertex distance.

Solution (a): With the y^2 term positive the asymptotes are $y = \pm \frac{a}{b}x$, where $a = \sqrt{25} = 5$ sits over y^2 and $b = \sqrt{144} = 12$ over x^2 . The rising slope is $\frac{5}{12}$

Note the orientation: for an up-down hyperbola the slope is a/b , not b/a .

Solution (b): The vertices are $\sqrt{25}$ from the centre. 5 cm

9. Garden bed 30 ft by 18 ft; distance between the foci.

Solution: The stated measurements are the full axes, so halve them first: $a = 15$ and $b = 9$.

$$c = \sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144} = 12$$

The sprinklers are $2c$ apart. 24 ft

Halving first is the whole problem here: using 30 and 18 directly gives a nonsense answer larger than the bed.

10. Gallery 40 m wide with foci 32 m apart. (a) Ceiling height. (b) Why wider foci means a lower ceiling.

Solution (a): Halve both given measurements: the semi-major axis is $a = 20$ and the focal distance is $c = 16$. For an ellipse $b^2 = a^2 - c^2$, and b is the height at the centre.

$$b = \sqrt{20^2 - 16^2} = \sqrt{400 - 256} = \sqrt{144}$$

12 m

Solution (b) — instructor-judged. The width is fixed, so $a = 20$ does not move, and the height comes from $b = \sqrt{a^2 - c^2}$. Pushing the listening spots apart increases c , which increases the amount being subtracted, so b falls. Geometrically: as the foci slide out toward

the ends of the ellipse, the curve has to flatten, and in the limit it collapses onto the segment joining them. Full credit connects a larger focal separation to a smaller height with a reason — either the algebraic chain or the flattening argument — rather than asserting it.
